

BUS 123 — Solving Business Problems with Technology

· Fall 2026

INTRO-M01-L03

Technology in Business

PRE-READING · Reading
Material Only

Pre-Reading

Technology in Business

Gerrish School of Business · BUS123

CONNECT TO PRIOR KNOWLEDGE

In the first two lessons of BUS123 you installed Microsoft 365, set up your OneDrive folder structure, and learned what this course is and how it works. You now have the tools and the expectations. This lesson asks a sharper question: why does any of it matter? Before you learn how to build a payroll model or analyze inventory data, it helps to understand the larger context — how technology actually functions inside a business, and what separates someone who uses it casually from someone who uses it professionally.

PASSIVE USER VS. PROFESSIONAL OPERATOR

Most people interact with technology every day without thinking about it deliberately. A passive user opens an app, does a task, and closes it. A professional operator thinks about what output the tool needs to produce, how to structure the work so others can verify it, and how to catch errors before they matter.

Dimension	Passive User	Professional Operator
Goal	Complete the task	Produce reliable output someone else can use
File management	Saves wherever is convenient	Follows a naming and folder convention every time
Formula use	Types answers directly into cells	References input cells so results update automatically
Error response	Deletes the error and retypes a number	Traces the formula to find and fix the root cause
Sharing	Emails an attachment	Shares a live OneDrive link to the current version

✓ The Shift This Course Is Asking You to Make

You do not need to be an expert to be a professional operator. You need to be deliberate. The habits you build in BUS123 -- naming files correctly, using formulas instead of typed values, checking your work before submitting -- are the same habits that make a junior analyst trustworthy to a manager.

TECHNOLOGY ACROSS BUSINESS SECTORS

Every industry uses technology differently, but the underlying skills are transferable. Understanding how technology shows up in the work you are heading toward helps you know which tools to prioritize.

Sector	Core Technology Problem	BUS123 Connection
Retail	Tracking inventory costs, pricing products, managing markdowns	COGS, markup/markdown, inventory math

Professional Services	Calculating employee compensation, modeling client loan scenarios	Payroll, TVM, interest calculations
Event Management	Projecting breakeven, managing seasonal cash gaps	Breakeven analysis, cash flow modeling
Healthcare	Analyzing payer mix, managing complex multi-rate payroll	Data analysis, payroll with variable rates
Finance	Modeling loan payments, evaluating investment returns	TVM, PMT, FV/PV functions

THE DIGITAL FLUENCY SPECTRUM

Employers think about digital skills in layers. Knowing where you currently sit on this spectrum -- and where you want to be -- helps you set a goal for what BUS123 can do for you.

Level	Description	What It Looks Like
1 — Basic User	Can open, navigate, and save files in standard applications	Types text in Word, opens spreadsheets, attaches files to email
2 — Functional User	Uses built-in tools to complete defined tasks reliably	Writes SUM formulas, formats a table, builds a basic chart
3 — Professional Operator	Structures work so it is accurate, auditable, and maintainable	Builds referenced formulas, names files correctly, checks for errors
4 — Analytical Producer	Uses technology to answer business questions and support decisions	Builds a payroll model, runs a breakeven analysis, interprets results

✓ Where BUS123 Targets You

Most students enter at Level 1 or 2. BUS123 is designed to move you to Level 3 by the end of the EXCEL track and to Level 4 by the end of the MATH track. Level 4 is where early-career business roles are won.

HOW TECHNOLOGY SHAPES DECISIONS

A spreadsheet does not just execute a calculation you already know how to do. It changes what questions you can even ask. Consider a retail buyer deciding how many units to order. Without a model, the decision is a guess. With a properly built spreadsheet, the buyer can run ten scenarios in two minutes and identify the order quantity that minimizes risk under each one.

This is the core reason BUS123 exists: technology does not replace business judgment -- it amplifies it. The analyst who can build the model gets to participate in the decision. The one who cannot has to wait for someone else's output.

WHERE AI FITS IN

AI tools, including tools like Microsoft Copilot built into Excel and Word, are increasingly present in business workflows. It is worth being honest about what they do and do not change.

AI is good at	AI is not good at
Drafting a first version of something	Knowing whether the answer is correct for your specific situation
Summarizing large amounts of text	Catching errors in a formula it did not write
Suggesting functions you did not know existed	Understanding the business context behind the numbers

■ The Skill AI Cannot Replace

AI tools are most valuable to people who already know how to do the work. If you cannot read a formula and judge whether it is correct, you cannot evaluate what AI produces. BUS123 builds the foundation that makes AI useful rather than dangerous.

TECHNOLOGY HABITS THAT PAY OFF

The research on early-career outcomes consistently shows that digital skill gaps are one of the most common reasons recent graduates underperform expectations. The habits below are simple -- but most students do not practice them until they learn the hard way.

■ Name every file correctly, every time

File naming is not a preference. It is a professional standard. An improperly named file signals carelessness before the content is even opened.

■ Save to the cloud, not the device

Local saves are one hardware failure away from lost work. Cloud saves are versioned, backed up, and accessible from anywhere.

■ Use formulas, not typed values

A typed number in a result cell is invisible to the next person who opens the file. A formula is a documented, auditable calculation.

■ Check your output before you share it

Sending a file with a formula error is worse than sending it late. Spend two minutes reviewing before submitting anything.

■ Learn the keyboard shortcut for what you do most

Speed compounds. Ten seconds saved per task adds up to hours across a semester -- and across a career.

KEY VOCABULARY

Passive User

Someone who uses technology reactively to complete a task without thinking deliberately about structure, accuracy, or how others will use the output.

Professional Operator

Someone who uses technology deliberately to produce reliable, auditable, and maintainable output that others can depend on.

Digital Fluency

The ability to use digital tools effectively and appropriately across a range of business contexts. Fluency implies understanding, not just task completion.

Digital Fluency Spectrum

A four-level framework describing how technology competence develops: from basic user through functional user, professional operator, to analytical producer.

Analytical Producer

The highest level of the digital fluency spectrum. An analytical producer uses technology to answer business questions, model scenarios, and support decisions rather than simply execute predefined tasks.

Formula Integrity

The property of a spreadsheet in which all calculated values are produced by formulas referencing input cells, making the model auditable, updateable, and free of hard-coded values in result cells.

Microsoft Copilot

An AI assistant built into Microsoft 365 applications including Excel and Word. Useful for drafting, summarizing, and suggesting functions -- but requires the user to evaluate and verify all output.

Payer Mix

In healthcare finance, the proportion of revenue coming from different payment sources such as private insurance, Medicare, Medicaid, and self-pay patients. Used in Harborside Medical Center case studies.